MTH2008 MATHEMATICAL ANALYSIS: SOLUTIONS TO JANUARY 2022 EXAM

SECTION A

1. (a) Let $S = (0, 1] \cup \{2\} \cup [3, 4)$.

(i) Is S open, closed, neither, or both? Justify your answer.

(ii) Find the complement S^c of S in $\mathbb{R}$.

(iii) Find the boundary ∂S of S.

(iv) Find the set of limit points of S. (10)

SOLUTION:

(i) S is neither open nor closed. It is not open since $1 \in S$ but there is no $\epsilon > 0$ such that the ϵ-neighbourhood of 1 is in S. It is not closed since its complement contains 0 but does not contain any ϵ-neighbourhood of 0. [4]

(ii) $S^c = (-\infty, 0] \cup (1, 2) \cup (2, 3) \cup [4, \infty)$. [2]

(iii) $\partial S = \{0, 1, 2, 3, 4\}$. [2]

(iv) The set of limit points of S is $[0, 1] \cup [3, 4]$. [2]

Similar to exercise

(b) Using the ϵ–δ definitions of limits and continuity, show that the function $f(x) = x^2 - x\sin(x)$ is continuous at $x_0 = 0$. (10)

SOLUTION:

By the definition of continuity, we must show that $\lim_{x\to 0} f(x) = f(0)$. Here $f(0) = 0$.

Thus, given $\epsilon > 0$, we must find a $\delta > 0$ so that $|f(x) - 0| < \epsilon$ when $0 < |x| < \delta$.

Provided that $\delta < 1$, we have

$$|f(x)-0| = |x^2-x\sin(x)| \leq |x^2|+|x\sin(x)| \leq |x|^2+|x| \leq 2|x| < 2\delta$$

whenever $|x| < \delta$. So if $\delta < \min(1, \frac{1}{2}\epsilon)$, we have $|f(x) - 0| < \epsilon$ when $|x| < \delta$. This shows that f is continuous at $x_0 = 0$. [10]

Similar to exercise

(c) Let $(a_n)_{n\geq 1}$ be a sequence of real numbers such that $a_n \to \infty$ as $n \to \infty$, and let $(b_n)_{n\geq 1}$ be an arbitrary sequence of real numbers. Prove that $a_n^2 + b_n^2 \to \infty$ as $n \to \infty$. (10)

SOLUTION: Given $M \in \mathbb{R}$, we need to show there is some $N \in \mathbb{N}$ such that $a_n^2 + b_n^2 > M$ for all $n > N$. Without loss of generality, we may assume $M > 1$. Since $a_n \to \infty$ as $n \to \infty$, there is some N such that $a_n > M$ for all $n > N$. But then

$$a_n^2 + b_n^2 > M^2 + b_n^2 > M^2 > M \text{ for all } n > N,$$

as required. [10]

Similar to exercise

(d) Show that the function $f : [0,1] \to \mathbb{R}$ given by $f(x) = 1 - x$ is Riemann integrable, and evaluate $\int_0^1 f(x)\,\mathrm{d}x$ using Riemann sums. (10)

SOLUTION: Let $P_n = (0, 1/n, 2/n, \ldots, 1)$ be the partition of $[0,1]$ into n equal parts. As $f(x)$ is decreasing, we have

$$\inf_{x\in[(i-1)/n,i/n]} f(x) = f\left(\frac{i}{n}\right)$$

and

$$\sup_{x\in[(i-1)/n,i/n]} f(x) = f\left(\frac{i-1}{n}\right).$$

So the lower Riemann sum is

$$\begin{aligned} L(f,P_n) &= \sum_{i=1}^{n} \left(\frac{i}{n} - \frac{(i-1)}{n}\right) \inf_{x\in[(i-1)/n,i/n]} f(x) \\ &= \frac{1}{n}\sum_{i=1}^{n} \left(1 - \left(\frac{i}{n}\right)\right) \\ &= 1 - \frac{1}{n^2}\sum_{i=1}^{n} i \\ &= 1 - \frac{1}{n^2}\cdot\frac{1}{2}n(n+1) \\ &= 1 - \frac{n+1}{2n} \\ &= \frac{n-1}{2n}. \end{aligned}$$

Similarly, the upper Riemann sum is

$$\begin{aligned} U(f,P_n) &= \sum_{i=1}^{n} \frac{1}{n} \sup_{x\in[2(i-1)/n,2i/n]} f(x) \\ &= \frac{1}{n}\sum_{i=1}^{n}\left(1-\frac{i-1}{n}\right) \\ &= \frac{1}{n}\left(n-\sum_{j=0}^{n-1}\frac{j}{n}\right) \\ &= 1-\frac{1}{n^2}\sum_{j=0}^{n-1} j \\ &= 1-\frac{1}{n^2}\cdot\frac{1}{2}\cdot(n-1)n \\ &= \frac{n+1}{2n}. \end{aligned}$$

Thus [6]

$$U(f,P_n)-L(f,P_n)=\frac{1}{n}\to 0 \text{ as } n\to 0.$$

Given $\epsilon > 0$, we can therefore choose a partition P_n so that $U(f,P_n) - L(f,P_n) < \epsilon$. Hence by Riemann's Integrability Criterion, f is integrable on $[0,1]$. [2]

[Alternatively, argue directly that $\sup_P(L(f,P)) \geq \inf_P(U(f,P))$, where the sup and inf are over all partitions.]

We have

$$\int_0^1 f(x)\,\mathrm{d}x = \lim_{n\to\infty} L(f,P_n) = \lim_{n\to\infty} U(f,P_n) = \frac{1}{2}.$$

[2]

Similar to exercise

(e) Find the derivative of the function $f:\mathbb{R}^2\to\mathbb{R}^2$ given by

$$f\big((x_1,x_2)^T\big) = \big(\sin(x_1)\cos(x_2),\cos(x_1)\sin(x_2)\big)^T.$$

(10)

SOLUTION:

The derivative is the linear map given (with respect to the standard basis of $\mathbb{R}^2$) by the matrix

$$\begin{pmatrix} D_1f_1 & D_2f_1 \\ D_1f_2 & D_2f_2 \end{pmatrix},$$

where f_1, f_2 are the first and second components of f:

$$f_1(x_1, x_2) = \sin(x_1)\cos(x_2), \qquad f_2(x_1, x_2) = \cos(x_1)\sin(x_2),$$

and D_i denotes partial differentiation with respect to x_i. Thus the derivative is the linear map represented by the matrix

$$\begin{pmatrix} \cos(x_1)\cos(x_2) & -\sin(x_1)\sin(x_2) \\ -\sin(x_1)\sin(x_2) & \cos(x_1)\cos(x_2) \end{pmatrix}.$$

[10]

Similar to exercise

[**50**]

SECTION B

2. (a) Let A and B be non-empty, bounded sets of real numbers, let $\alpha = \sup(A)$ and $\beta = \sup(B)$, and suppose that $\alpha < \beta$.

Determine whether each of the following statements is necessarily true, or can be false. Justify your answers with a proof or a counterexample.

(i) $a \leq b$ for all $a \in A$ and $b \in B$;

(ii) there exists an element $b \in B$ such that $b \notin A$;

(iii) if $\beta \notin B$ then B contains infinitely many distinct elements which are upper bounds for A. (15)

SOLUTION:

(i) This can be **FALSE**. For example take $A = (0, 3)$ and $B = (0, 4)$, and let $a = 2 \in A$ and $b = 1 \in B$. Then $\alpha = 3$ and $\beta = 4$, so $\alpha < \beta$, but $a > b$. [3]

(ii) This is **TRUE**. As $\alpha < \beta$, we know that α is not an upper bound for B, so there is some $b \in B$ with $b > \alpha$. As α is an upper bound of A, we then have $b \notin A$. [5]

(iii) This is **TRUE**. By (ii), we can find $b_1 \in B$ with $\alpha < b_1$. Then $b_1 \leq \beta$, but $b_1 \neq \beta$ since $\beta \notin B$. Hence $\alpha < b_1 < \beta$. Suppose inductively we have found elements $b_1, \ldots, b_n \in B$ with $\alpha < b_1 < \cdots < b_n < \beta$. As b_n is not an upper bound for B, there is some $b_{n+1} \in B$ with $b_n < b_{n+1}$. Then $b_{n+1} < \beta$ since β is an upper bound for B and $\beta \notin B$. Thus we obtain an infinite sequence $(b_n)_{n \geq 1}$ of distinct elements of B, and each b_n is an upper bound for A since $b_n > \alpha$. [7]

Similar simple properties of sup, limit points, etc, seen as exercises, but these specific statements unseen. (i) and (ii) should be easy; (iii) more challenging.

(b) For $n \geq 2$, let $f_n : [0, 1] \to \mathbb{R}$ be the function given by

$$f_n(x) = \begin{cases} 1 - nx & \text{if } 0 \leq x \leq \dfrac{1}{n}, \\ \dfrac{nx - 1}{n - 1} & \text{if } \dfrac{1}{n} < x \leq 1. \end{cases}$$

Find the pointwise limit f of the sequence of functions $(f_n)_{n \geq 2}$. Does f_n converge to f uniformly on $[0, 1]$? Justify your answer. (10)

SOLUTION:

For $0 < x \leq 1$, if $n > \frac{1}{x}$ we have

$$f_n(x) = \frac{nx - 1}{n - 1},$$

so $f_n(x) \to x$ as $n \to \infty$. If $x = 0$, we have $f_n(x) = 1$ for all n. Hence the pointwise limit of f_n is f where

$$f(x) = \begin{cases} 1 & \text{if } x = 0; \\ x & \text{if } 0 < x \leq 1. \end{cases}$$

[5]

f_n does not converge to f uniformly. Given $n \geq 2$, let $x = \frac{1}{2n}$. Then

$$f_n(x) = 1 - n \cdot \frac{1}{2n} = \frac{1}{2}$$

but $f(x) = x = \frac{1}{2n}$. So $|f_n(x) - f(x)| = \frac{1}{2} - \frac{1}{2n} \geq \frac{1}{4}$. Thus there is no N such that $|f_n(x) - f(x)| < \frac{1}{4}$ for all $n > N$ and all $x \in [0, 1]$. This says that the convergence of f_n to f is not uniform.

[Alternative: Note that f_n is continuous for all n and that f is not continuous at 0, and quote the fact that a uniform limit of continuous functions is continuous.] [5]

Similar to exercises and example in lecture notes.

3. (a) Let $f : [a, b] \to \mathbb{R}$ be a continuous function such that $f(a) > 0$ and $f(b) < 0$, and let

$$S = \{x \in [a, b] : f(x) < 0\}.$$

Show that the set S has an infimum (greatest lower bound) c, and prove that $f(c) = 0$. (12)

SOLUTION: S is non-empty since $b \in S$, and S is bounded below by a. Thus S has an infimum by completeness of $\mathbb{R}$. [3]

Let $c = \inf(S)$, and suppose that $f(c) > 0$. Then $c \neq b$, so f is continuous from the right at c. (We cannot simply say that f is continuous at c unless we first show $c \neq a$.) Thus there is a $\delta > 0$ such that if $c \leq x < c + \delta$ then $x \in (a, b)$ and $|f(x) - f(c)| > f(c)$. For these x, we have $f(x) > 0$. For $a \leq x \leq c$ we have $x \notin S$ since c is a lower bound for S, and so $f(x) \geq 0$ for these x. Thus $f(x) \geq 0$ for all $x < c + \delta$, so $c + \delta$ is a lower bound for S. This contradicts the definition of c as the *greatest* lower bound for S. [4]

Suppose $f(c) < 0$. Then $c \neq a$ and f is continuous from the left at c. So there is a $\delta > 0$ such that if $c - \delta < x \leq c$ then $x \in (a, b)$ and $|f(x) - f(c)| < |f(c)|$. For these x we have $f(x) < 0$. So there are $x < c$ with $x \in S$, showing that c is not a lower bound for S. This contradicts the definition of c. [4]

As the conditions $f(c) > 0$ and $f(c) < 0$ each lead to a contradiction, we have $f(c) = 0$ as required [1]

Unseen in this form: this is essentially the proof of the Intermediate Value Theorem, but set up differently to what is in the lecture notes.

(b) Let $g : [a, b] \to \mathbb{R}$ be a bounded function, and let $g^+ : [a, b] \to \mathbb{R}$ be defined by

$$g^+(x) = \begin{cases} g(x) & \text{if } g(x) \geq 0, \\ 0 & \text{if } g(x) < 0. \end{cases}$$

(i) Show that, for any closed interval $[c, d] \subseteq [a, b]$, we have

$$0 \leq \sup_{x \in [c,d]} g^+(x) - \inf_{x \in [c,d]} g^+(x) \leq \sup_{x \in [c,d]} g(x) - \inf_{x \in [c,d]} g(x).$$

(ii) Deduce that if g is Riemann integrable on $[a, b]$ then g^+ is also Riemann integrable on $[a, b]$. (13)

SOLUTION:

(i) Let $[c,d] \subseteq [a,b]$. For all $x \in [c,d]$, we have

$$\inf_{x\in[c,d]} g^+(x) \leq g^+(x) \leq \sup_{x\in[c,d]} g^+(x)$$

so that

$$0 \leq \sup_{x\in[c,d]} g^+(x) - \inf_{x\in[c,d]} g^+(x)$$

(and similarly for g in place of g^+).
For the second inequality, consider the following cases:

- if $g(x) \geq 0$ for all $x \in [c,d]$ then $g^+(x) = g(x)$ for all $x \in [c,d]$, so

$$\sup_{x\in[c,d]} g^+(x) - \inf_{x\in[c,d]} g^+(x) = \sup_{x\in[c,d]} g(x) - \inf_{x\in[c,d]} g(x).$$

- if $g(x) \leq 0$ for all $x \in [c,d]$ then $g^+(x) = 0$ for all $x \in [c,d]$, so that

$$\sup_{x\in[c,d]} g^+(x) - \inf_{x\in[c,d]} g^+(x) = 0 \leq \sup_{x\in[c,d]} g(x) - \inf_{x\in[c,d]} g(x).$$

- if g changes sign on $[c,d]$, so there exist x_+, $x_- \in [c,d]$ with $g(x_1) > 0$ and $g(x_2) < 0$, then

$$\sup_{x\in[c,d]} g^+(x) = \sup_{x\in[c,d]} g(x) \text{ and } \inf_{x\in[c,d]} g(x) < 0 = \inf_{x\in[c,d]} g^+(x),$$

so that

$$\sup_{x\in[c,d]} g^+(x) - \inf_{x\in[c,d]} g^+(x) = \sup_{x\in[c,d]} g(x) \leq \sup_{x\in[c,d]} g(x) - \inf_{x\in[c,d]} g(x).$$

These three cases cover all possibilities, so the required inequality holds in all cases. [7]

(ii) Suppose that g is Riemann integrable on $[a,b]$. By Riemann's Integrability Criterion, given $\epsilon > 0$ there exists a partition $P = (a = x_0, x_1, \ldots, x_n = b)$ of $[a,b]$ such that

$$U(g,P) - L(g,P) < \epsilon.$$

Thus we have

$$\sum_{i=1}^{n} (x_i - x_{i-1}) \left(\sup_{x\in[x_{i-1},x_i]} g(x) - \inf_{x\in[x_{i-1},x_i]} g(x) \right) < \epsilon.$$

Now applying (i) to each interval $[x_{i-1}, x_i]$ we have

$$0 \leq \sup_{x\in[x_{i-1},x_i]} g^+(x) - \inf_{x\in[x_{i-1},x_i]} g^+(x) \leq \sup_{x\in[x_{i-1},x_i]} g(x) - \inf_{x\in[x_{i-1},x_i]} g(x)$$

for $1 \leq i \leq n$. Substituting into the previous inequality gives

$$\sum_{i=1}^{n}(x_i - x_{i-1})\left(\sup_{x\in[x_{i-1},x_i]} g^+(x) - \inf_{x\in[x_{i-1},x_i]} g^+(x)\right) < \epsilon,$$

so that

$$U(g^+, P) - L(g^+, P) < \epsilon.$$

Using Riemann's Integrability Criterion again, it follows that g^+ is Riemann integrable on $[a, b]$. [6]

Unseen, but similar result that g integrable implies $|g|$ integrable seen as exercise.) **[25]**

4. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined at $x = (x_1, x_2)^T$ by

$$f((x_1, x_2)^T) = \begin{cases} \dfrac{2x_1^2 + x_2^2}{x_1^2 + 2x_2^2} & \text{if } x \neq (0,0)^T, \\ 1 & \text{if } x = (0,0)^T. \end{cases}$$

(a) Determine whether f is continuous at $(0,0)^T$. (5)

SOLUTION:

Given $\delta > 0$, the point $y = (\frac{1}{2}\delta, 0)^T$ satisfies $||y|| < \delta$ and $f(y) = 2$, so that $||f(y) - f(\mathbf{0})|| = 1$ (where $\mathbf{0} = (0,0)^T$). Hence, choosing $\epsilon = 1$, there is no $\delta > 0$ such that $||f(x) - f(\mathbf{0})|| < \epsilon$ for all x with $||x - \mathbf{0}|| < \delta$. This shows that f is **not** continuous at $\mathbf{0}$. [5]

Seen similar in lectures, but the idea is a bit subtle. Note that I am **NOT** asking them to show the singularity is essential.

(b) Show that, for each point $(p_1, p_2) \neq (0,0)$, f is differentiable at (p_1, p_2), and find its derivative there. (10)

SOLUTION:

At any point $(x_1, x_2) \neq (0,0)^T$, the Jacobian of f is

$$\left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right)$$

where

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= \frac{(x_1^2 + 2x_2^2)(4x_1) - (2x_1^2 + x_2^2)(2x_1)}{(x_1^2 + 2x_2^2)^2} \\ &= \frac{6x_1 x_2^2}{(x_1^2 + 2x_2^2)^2} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial f}{\partial x_2} &= \frac{(x_1^2 + 2x_2^2)(2x_2) - (2x_1^2 + x_2^2)(4x_2)}{(x_1^2 + 2x_2^2)^2} \\ &= \frac{-6x_1^2 x_2}{(x_1^2 + 2x_2^2)^2} \end{aligned}$$

[7]

These partial derivatives are continuous at all points $(x_1, x_2) \neq (0,0)^T$. Hence f is differentiable at (p_1, p_2), and its derivative there is the linear map with matrix

$$\left(\frac{6p_1 p_2^2}{(p_1^2 + 2p_2^2)^2}, \frac{-6p_1^2 p_2}{(p_1^2 + 2p_2^2)^2} \right).$$

[3]

Routine and similar to exercises

(c) Find all the stationary points of f. (5)

SOLUTION:

Both partial derivatives vanish if and only if $6x_1x_2^2 = -6x_1^2x_2 = 0$, which holds if and only if either $x_1 = 0$ or $x_2 = 0$. Thus the stationary points are

$$\{(0, x_2)^T : x_1 \neq 0\} \cup (x_1, 0) : x_1 \neq 0\}.$$

[Note that $(0,0)^T$ is not a stationary point since f is not continuous, and hence not differentiable, there.] [5]

"Unseen" in that students will have to figure out how to write the union of two lines, and notice that the origin is excluded. They will have seen simpler examples of finding stationary points.

(d) Does f have a global minimum? Justify your answer. (5)

SOLUTION:

f has global minimum $\frac{1}{2}$. This value is attained at any point with $x_1 = 0$ and $x_2 \neq 0$, and is clearly less than the value 1 at $(0,0)^T$. At any point other than $(0,0)^T$, we have

$$\begin{aligned} f((x_1, x_2)^T) - \frac{1}{2} &= \frac{2x_1^2 + x_2^2}{x_1^2 + 2x_2^2} - \frac{1}{2} \\ &= \frac{(4x_1^2 + 2x_2^2) - (x_1^2 + 2x_2^2)}{x_1^2 + 2x^2} \\ &= \frac{3x_1^2}{x_1^2 + 2x_2^2} \\ &\geq 0. \end{aligned}$$

[5]

Unseen.

[**25**]